Mehrzad Golabi

M.Sc. Student in Control Engineering, Iran University of Science and Technology · Tehran, Iran

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Research Interests

M.Sc. student in control engineering with a background in mobile robotics, FPGA hardware design and industrial automation, interested in research that leads to building and deploying real systems, especially:

- Localization and state estimation for mobile robots (SLAM and sensor fusion)
- Real-time control implementation on FPGAs and embedded systems
- Industrial automation and digital twins

Education

M.Sc. in Electrical Engineering (Control Systems), Iran University of Science and Technology (IUST), Tehran — Sep 2026 – present

B.Sc. in Electrical Engineering (Control Systems), Shahid Beheshti University, Tehran — Sep 2021 – Feb 2026

GPA: 15.69 / 20 · Last three semesters: 18.60 / 20

Relevant courses (grade out of 20): Embedded Systems (20), Microprocessor Systems (20), Digital Systems II (20), Industrial Control (20), FPGA-Based Digital System Design (19.5), Industrial Automation (19.5), Advanced Programming (19), Instrumentation (18.75), Linear Control (18), Digital Control (17.25)

Laboratories: Digital Systems II (20), Instrumentation (20), Digital Control (19.5), Electrical Drives (19.5), Industrial Control (19.25), Linear Control (19)

B.Sc. thesis (supervisor: Dr. Mostafa Abedi; grade 20 / 20): *Comparison of Simultaneous Localization and Mapping (SLAM) Algorithms on the TurtleBot3 Robot*

- Implemented Hector SLAM from scratch in MATLAB (multi-resolution Gauss–Newton scan matching, bilinear interpolation, log-odds occupancy map), publishing the map and TF to ROS 2
- Developed a pose-graph SLAM that removes transient obstacles from the map dynamically (Bresenham ray tracing) and automatically rolls back false loop closures
- Evaluated and compared both algorithms in Gazebo and in four real-world scenarios on a TurtleBot3 Burger with an LDS-02 LiDAR running ROS 2 Humble
- Collaborated with the university robotics team on the robot's autonomous navigation system

Research Experience

Course paper, Industrial Automation: *Digital Twins in Industrial Automation in the Iranian Industrial Context* — 2025

- Reviewed multi-layer digital-twin architectures, IIoT technologies and industrial protocols (Modbus, Profibus, OPC UA, MQTT)
- Analysed brownfield equipment retrofitting, edge-first architecture and a low-cost “good-enough twin” approach for Iranian industry

Teaching and Work Experience

Teaching Assistant, Shahid Beheshti University

- Industrial Control — Feb 2025 – Feb 2026 (2 semesters)
- Modern Control (MATLAB and Simulink instruction) — Sep 2025 – Feb 2026 (1 semester)
  github.com/MehrzadGolabi/Modern_Control
- Power Systems Analysis I — Feb 2025 – Sep 2025 (1 semester)

Industrial Control Laboratory Supervisor, Shahid Beheshti University — Sep 2025 – Feb 2026 (1 semester)

Nasb Niroo Iran — Summer 2025 – present

- Review and correct P&ID drawings and validate engineering data

- R&D internship (summer 2025): reviewed and completed technical documentation for turbines and related components
- Drafted instrumentation P&ID drawings in AutoCAD Plant 3D

Projects

Line Follower Robot (ROS 2 Jazzy) Dec 2025

- Designed and built a line-following mobile robot in ROS 2 Jazzy with two modes: a five-sensor IR array and camera-based image processing
- Modelled the robot and Raspberry Pi camera in URDF/SDF and built line-following tracks in Gazebo
- Structured the software as three packages (bringup, machine vision, IR) with nodes, topics, services and parameterised launch files for both the real robot and simulation
- Implemented line detection with OpenCV and a state-machine follower for the IR sensors
- Drove motors and encoders through ros2_control and arbitrated motion commands with twist_mux
- Bridged Gazebo to ROS 2 with ros_gz_bridge, running the same nodes in simulation (SITL) and on hardware (HITL); debugged with RViz2 and rqt

  github.com/MehrzadGolabi/ROS2_linefollower

Reaction-Wheel Inverted Pendulum Sep 2025

- Designed and built a reaction-wheel inverted pendulum for the Digital Control course (3D-printed body, Arduino Nano, TB6612 driver, AS5600 encoder)
- Identified the DC motor with MATLAB's System Identification Toolbox (three-pole model, over 95% fit)
- Discretised a PID controller with the Tustin method in Simulink and implemented it on the microcontroller at a 20 Hz sampling rate

FPGA-Based Ball and Beam System Jun 2025

- Implemented a PID controller in VHDL on a Xilinx Spartan-6 FPGA to control ball position using a servo motor and an ultrasonic sensor

  github.com/MehrzadGolabi/Ball_and_Beam_VHDL

FPGA-Based AES Encryptor Jun 2025

- Implemented the AES encryption algorithm in hardware (VHDL) on an FPGA and built a Python command-line interface for it

  github.com/MehrzadGolabi/AES-VHDL

Elevator Control System Jan 2025

- Designed and built a model elevator and implemented its sequential control system on a Raspberry Pi Pico microcontroller

  github.com/MehrzadGolabi/Elevator_Project

Self-Balancing Robot Jul 2024

- Built a self-balancing robot on an ESP32 microcontroller with closed-loop balance control

  github.com/MehrzadGolabi/selfbalancingrobot

Skills

- **Programming and HDL:** C, C++, Python, MATLAB, Assembly (AVR/ARM), VHDL, Bash
- **Robotics:** ROS 2 (Humble, Jazzy), ros2_control, tf2, Nav2, micro-ROS; SLAM (pose-graph, Hector)
- **Simulation and modelling:** Gazebo Sim, URDF/SDF, ros_gz_bridge, RViz2, rqt; MATLAB/Simulink, System Identification Toolbox
- **Embedded systems and circuit design:** ESP32, STM32, Raspberry Pi 5, Raspberry Pi Pico, Arduino, Xilinx Spartan-6 FPGA; STM32CubeIDE, Proteus, Altium Designer, LTspice, Xilinx ISE
- **Industrial automation and drafting:** Siemens PLC (TIA Portal, STEP 7, LOGO! Soft Comfort); industrial networks (Modbus, Profibus); AutoCAD, AutoCAD Plant 3D (P&ID)
- **Tools and mechanical design:** Linux, Git, Docker, OpenSCAD (parametric 3D design), 3D printing

Languages

Persian: native · **English:** professional working proficiency (IELTS Academic to be taken in winter 2026–27)